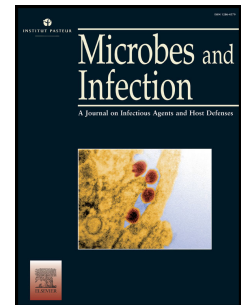


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**Combination of colistin and tobramycin inhibits persistence of *Acinetobacter baumannii*
by membrane hyperpolarization and down-regulation of efflux pumps**

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Abstract

Acinetobacter baumannii, a leading cause of nosocomial infections, is a serious health threat. Limited therapeutic options due to multi-drug resistance and tolerance due to persister cells have urged the scientific community to develop new strategies to combat infections caused by this pathogen effectively. Since combination antibiotic therapy is an attractive strategy, the effect of combinations of antibiotics, belonging to four classes, was investigated on eradication of persister cells in *A. baumannii*. Among the antibiotics included in the study, tobramycin-based combinations were found to be the most effective. Tobramycin, in combination with colistin or ciprofloxacin, eradicated persister cells in *A. baumannii* in late exponential and stationary phases of growth. Mechanistically, colistin facilitated the entry of tobramycin into cells by increasing membrane permeability and inducing hyperpolarization of the inner membrane accompanied by increase in ROS production. Expression of the genes encoding universal stress protein and efflux pumps was down-regulated in response to tobramycin and colistin, suggesting increased lethality of their combination that might be responsible for eradication of persister cells. Thus, a combination of tobramycin and colistin could be explored as a promising option for preventing the relapse of *A. baumannii* infections due to persister cells.

Keywords: *Acinetobacter baumannii*; persister; combination therapy; colistin; tobramycin

1. Introduction

Acinetobacter baumannii is an aerobic, non-fermentative, Gram-negative, opportunistic pathogen responsible for a variety of hospital-acquired infections, including hospital-acquired pneumonia (HAP), ventilator-associated pneumonia (VAP), urinary tract infections, bacteremia, meningitis, skin and soft tissue infections, especially among patients with compromised immune system [1]. *A. baumannii* has a remarkable ability to persist in the hospital environment and acquire resistance to all currently used antimicrobial agents [2,3]. In 2017, World Health Organization (WHO) published antibiotic-resistant ‘priority pathogens’ list with *A. baumannii* at the top in the “critical” category marked for the development of new antibiotics [4].

Multidrug-resistant (MDR) strains of *A. baumannii* are continuously evolving for which novel treatment approaches are desperately needed but equally worrying situation is to control chronic infection caused by antibiotic sensitive strains that form persister cells, a sub-population of non-growing, metabolically dormant cells that resist killing by lethal concentrations of bactericidal antibiotics [5,6]. These persister cells are responsible for the treatment failure in chronic and biofilm-associated infections and have been identified as a possible reservoir for the development of resistance [7]. These phenotypic variants have reduced metabolic activity and are capable of restoring the bacterial population as soon as the antibiotic pressure drops, leading to the recurrence of infection [8,9]. Persister cells are genetically distinct from antibiotic-resistant cells but identical (isogenic) to antibiotic susceptible wild type and their tolerance to antibiotics is transient and non-inheritable [10,11]. *A. baumannii* forms persister cells in response to antibiotics belonging to β -lactam group [12], aminoglycosides [13,14], and polymyxin [15]. Ineffectiveness of monotherapies with

antibiotic has made it imperative to explore effective antibiotic combinations to combat *A. baumannii* infections.

As combination therapy has been successfully used for the treatment of tuberculosis, viral infections, and many types of cancers [16], reports are now available where colistin-based antibiotic combinations have been effective in eradicating persister cells formed by *A. baumannii* and *Pseudomonas aeruginosa* [17,18]. The combination of polymyxin B and meropenem inhibited persister cells in *A. baumannii* in the late exponential phase of growth after exposure for 48 h [19]. In another study, combination of colistin and amikacin effectively eradicated *A. baumannii* persister cells in early exponential phase cells [18]. Both studies highlighted the synergistic bactericidal effect of drug combinations but did not explain the possible mechanism(s) responsible for the eradication of persister cells.

In this context, the effect of different antibiotic combinations on persister cells in *A. baumannii* was evaluated. Tobramycin-based combinations significantly decreased the persistence of *A. baumannii*. Tobramycin in combination with colistin was the most effective as persister cells were completely eradicated after the treatment. Eradication of persister cells could be due to the increased membrane permeability, membrane potential and ROS production by colistin, and inhibition of efflux pumps by these antibiotics.

2. Material and methods

2.1 Bacterial strains and culture conditions

The strains of *A. baumannii* ATCC 17978 and *A. nosocomialis* M2 [20] were used in this work. All experiments were conducted at 37°C in LB broth with shaking at 180 rpm.

2.2 Determination of Minimum Inhibitory Concentration (MIC)

MIC was determined by the broth microdilution method according to CLSI guidelines [21,22]. Antibiotics were serially diluted in a range from 128 to 0.25 µg/ml. The concentration where there was no visible growth was taken as MIC.

2.3 Persister assays

A single colony was grown in LB broth at 37 °C for 16 h. For the late exponential phase, the overnight grown culture was diluted 1:100 with fresh LB broth and grown for 4 h at 180 rpm. For the stationary phase, the overnight grown culture was diluted 1:100 with LB broth and grown to the early exponential phase ($OD_{600} \sim 0.3$) at 37 °C to obtain cells having synchronized growth. This was further diluted 1:1000 with LB broth and incubated for 16 h at 37 °C, 180 rpm [23].

Concentration-dependent persister assay was performed with cells from a specific growth phase treated with different concentrations of antibiotics ($X \text{ MIC}$) for 3 h [24]. The time-dependent persister assay was performed with fixed concentrations of antibiotics as selected from the concentration-dependent assay (falling in the plateau region reflecting the persister population), for different (3 to 24 h) time intervals. At different time points after the addition of antibiotics, cells were harvested by centrifugation at $8000 \times g$ for 10 min, washed twice with phosphate buffer saline (PBS), and diluted 10-fold serially in PBS. After that 10 µl

aliquot of each dilution was spot plated on LB agar plates for colony-forming units per ml (CFU/ML) [25]. Only dilutions that yielded 10–100 colonies were counted and percentage survival was determined.

2.4 Outer membrane permeability assay

Late exponential phase cells (1.0 ml) were treated with colistin, tobramycin or their combination for 1 h at 37 °C, 180 rpm, washed twice and re-suspended in 1.0 ml PBS (pH 7.4). 200 µl of the cell suspension was taken in a black microtiter plate to which 10 µM N-phenyl-1-naphthylamine (NPN) was added. NPN fluorescence was determined immediately using a multimode microplate reader (BioTek) with excitation and emission wavelengths of 350 and 420 nm, respectively [26]. Fluorescence was normalized with growth (OD₆₀₀) of the respective samples. Untreated cells were taken as the control.

2.5 Membrane potential measurement

Late exponential phase cells were harvested, washed and re-suspended in PBS containing 1% glucose. The cell suspension was treated with 10 µM DiOC₂ (3,3-diethyloxacarbocyanine iodide) for 30 min at 37 °C, 180 rpm in the dark, and then transferred to Genetix 96-well black microtiter plate. The cells were analyzed using multimode microplate reader (BioTek) with excitation and emission wavelengths set at 485 and 650 nm respectively. After establishing a baseline reading, colistin, tobramycin or their combination was added to the wells, and the change in red fluorescence was monitored over time [27].

2.6 Reactive oxygen species (ROS) determination

Late exponential phase cells were harvested, washed, and re-suspended in PBS. The cell suspension was treated with colistin, tobramycin, or their combination for 1 h at 37 °C, 180

rpm. 5 μ M of 2',7'-dichlorofluorescein diacetate (DCFDA) was added to the cell suspension and incubated in the dark at 37 °C for 30 min [28]. The fluorescence was measured using a microplate reader with excitation and emission wavelengths set at 488 and 530 nm, respectively. The fluorescence intensity (FI)/OD₆₀₀ was calculated for normalizing the fluorescence with respect to growth. Untreated cells were processed similarly and were used as control.

2.7 Gene Expression Quantification by RT- qPCR

Total RNA was isolated using TRI Reagent (Sigma) according to the manufacturer's instructions. Late exponential phase cells (1.0 ml) of *A. baumannii* ATCC 17978 were treated with colistin (10X) and tobramycin (40X) for 30 min at 37 °C, 180 rpm. Untreated cells were used as control. After the treatment, cells were centrifuged at 1000 $\times$ g for 10 min and the supernatant was decanted carefully to remove ghost cells and debris from the sample [29]. The cell pellet was washed twice with PBS (pH 7.4) and centrifuged at 1000 $\times$ g for 5 min to obtain unlysed cells. RNA was converted to cDNA using the Verso cDNA kit (Thermo Scientific). The expression of genes was assessed using iTaq Universal SYBR Green Supermix (Biorad) and CFX96 Touch Real-Time PCR (Biorad). The qRT-PCR primers were designed using the Primer 3 software (Table S1) with 16S rRNA gene as the internal control. For each gene tested, three biological replicates and two technical replicates were performed and the relative gene expression was calculated by the $2^{-\Delta\Delta Ct}$ method [30].

3. Results

3.1 Effect of antibiotic combinations on persister cells

Concentration-dependent assay of late exponential phase *A. baumannii* 17978 culture with rifampicin (MIC, 4 µg/mL), ciprofloxacin (MIC, 0.25 µg/mL), tobramycin and colistin (MIC, 2 µg/mL each) showed biphasic killing curves, typical of persister cells (Fig. S1). The time-dependent assay revealed the presence of maximum persister cells on treatment with rifampicin (4.5% at 40X MIC) followed by ciprofloxacin (0.17% at 20X MIC), colistin (0.011% at 10X MIC), and least on treatment with tobramycin (0.0001% at 40X MIC) for 24 h (Fig. 1).

The effect of paired antibiotics on persister cells was investigated by combining tobramycin/colistin, tobramycin/ciprofloxacin, tobramycin/rifampicin, colistin/ ciprofloxacin, colistin/rifampicin and ciprofloxacin/rifampicin. Treatment of late exponential phase cells with tobramycin in combination with colistin or ciprofloxacin eradicated the persister cells in 3 h. Combination of tobramycin with rifampicin significantly reduced persister cells by 3 log-fold as compared to rifampicin alone at 24 h (Fig. 1A). On the contrary, other antibiotic combinations viz. colistin/ciprofloxacin, colistin/rifampicin, and ciprofloxacin/rifampicin showed no significant decrease in persister cells as compared to monotherapy alone (Fig. 1B). Similar results were observed against *A. nosocomialis* strain M2, which is genetically closely related and phenotypically similar to *A. baumannii*. *A. nosocomialis* M2 treated with 40X tobramycin formed 0.01% persisters (MIC, 4 µg/mL) and 0.001% against 40X colistin (MIC, 2 µg/mL) but tobramycin-colistin combination eradicated the persister cells (Fig. S2).

Since persister cells are known to form higher fraction in stationary phase culture, tobramycin-based combinations were evaluated for their effect on persister cells in stationary

phase as well. Concentration-dependent assay of stationary phase *A. baumannii* 17978 with rifampicin, ciprofloxacin, colistin and tobramycin also showed biphasic killing curves (Fig. S3). Time-dependent assay with stationary phase *A. baumannii* cells showed maximum persisters with rifampicin (5.4% at 60X MIC) followed by ciprofloxacin (4.5% at 60X MIC), colistin (0.0012% at 60X MIC), and least with tobramycin (0.0003% at 80X MIC) at 24 h (Fig. S4). Treatment with combination of tobramycin and colistin for 3 h eradicated persister cells in *A. baumannii* stationary phase culture. Tobramycin in combination with ciprofloxacin or rifampicin significantly decreased persister cells by 2.5- and 4 log-fold respectively as compared to monotherapy of stationary phase cells for 24 h (Fig. S4).

3.2 Change in membrane permeability

To elucidate the mechanism of persister cells eradication by colistin and tobramycin, change in the outer membrane permeability of *A. baumannii* cells was measured using NPN fluorescence. Treatment of late exponential phase cells (3.2×10^3 CFU/ml) that remained after treatment with combination of tobramycin and colistin for 1 h showed a 7.65-fold ($P \leq 0.001$) increase in NPN fluorescence as compared to the untreated cells (3.3×10^9 CFU/ml), indicating increase in membrane permeability. Treatment of late exponential phase cells with 10X colistin alone showed a significant increase in NPN fluorescence by 8.3-fold ($P \leq 0.01$) while there was no significant change in the fluorescence intensity of NPN after treatment with 40X tobramycin (Fig. 2).

3.3 Change in membrane potential

The effect of tobramycin and colistin on the inner membrane of *A. baumannii* was studied by using DiOC₂, a membrane potential-sensitive dye. At low concentration, DiOC₂ exhibits green fluorescence in bacteria, however, fluorescence emission shifts towards red as the dye

becomes more concentrated due to self-association caused by the increased membrane potential. Treatment of late exponential phase cells with the combination of tobramycin with colistin showed significant increase in red fluorescent intensity by 2.3-fold ($P \leq 0.05$) indicating hyperpolarization of the cell membrane. Treatment with 10X colistin showed a significant increase in this red fluorescence by 2.7-fold ($P \leq 0.01$). While, upon the treatment of *A. baumannii* cells with 40X tobramycin, there was significant decrease in red fluorescence by 10.9-fold (Fig. 3).

3.4 ROS generation

To evaluate the production of ROS, DCFDA was used as a fluorophore, which gets converted into strong fluorescent 2',7'-dichlorofluorescein (DCF) in the presence of ROS. Treatment of late exponential phase cells with a combination of tobramycin and colistin induced a significant increase in ROS levels by 9.14-fold ($P \leq 0.01$). Colistin alone was found to significantly increase ROS levels by 8.2-fold ($P \leq 0.05$). However, treatment with tobramycin showed insignificant change in ROS production (Fig. 4).

3.5 Expression of genes encoding universal stress protein, isocitrate lyase and efflux pumps

Since no cell survival was observed after treatment for 3 h with combination of tobramycin and colistin, the expression of genes was determined in response to treatment with either tobramycin or colistin individually after 30 min treatment with 2.0×10^7 and 2.0×10^8 CFU/ml respectively. The expression of universal stress protein *uspA* was significantly down-regulated by 2.2- and 2-fold after treatment with tobramycin and colistin respectively. Similarly, treatment with tobramycin showed significant down-regulation (2.5-fold) in the expression of *icl*, a key enzyme of glyoxylate shunt. However, colistin significantly increased its expression by 2.2-fold. Treatment with tobramycin decreased *adeB*, *emrA*, and *emrB*

expression by 3.8-, 4- and 3.5-fold respectively. Colistin also showed significant down-regulation of *adeB*, *emrA*, and *emrB* expression by 2.2-, 3.9- and 2.9-fold respectively (Fig. 5).

4. Discussion

A. baumannii has become the pathogen of major global public health concern [4] due to its ability to develop multidrug resistance. It has been postulated that persistence is an important contributor to the emergence of resistance and is responsible for the recurrence of infections. In this study, we found that *A. baumannii* showed varied persistence to different classes of antibiotics and the number of persister cells also varied with the phase of growth as has been reported earlier [13,31]. Among all the antibiotics tested, tobramycin was the most effective, as least number of persister cells were detected in its presence both in late log and stationary phase cultures. Since currently used antimicrobial monotherapies are not effective in eradicating persister cells, combination therapy based on antibiotics with contrasting mechanisms of action could be used to combat persister cells. Simultaneous treatment with tobramycin in the presence of colistin or ciprofloxacin eradicated persister cells in the late exponential and stationary phase *A. baumannii* cultures. Previous studies have shown combination of colistin-amikacin and polymyxin B-meropenem to be effective in eradicating *A. baumannii* persister cells [18] but the mechanism involved has not been explained.

The outer membrane of Gram-negative bacteria is a major barrier to the uptake of antibiotics due to the presence of lipopolysaccharides [32] and is also crucial for the viability of persister cells. Hence, membrane has been suggested as a potential target for anti-persister strategies [33]. The combination of colistin with tobramycin showed increase in the permeability of outer membrane of *A. baumannii* cells. Previously we have shown the disruption of

membrane integrity of *A. baumannii* by colistin through FESEM [31]. Colistin directly damages the outer membrane of Gram-negative bacteria by interfering with LPS bridging interactions [34].

Alteration in membrane potential is considered an early indicator of injury in bacteria. The combination of tobramycin with colistin showed membrane hyperpolarization of *A. baumannii* cells, induced by colistin. Hyperpolarization occurs primarily due to the change in hydrogen ion concentration or by increased movement of K^+ ions that diffuse out of the plasma membrane through K^+ channels and affect cellular equilibrium [35]. A recent study has shown that increased magnesium flux prevented hyperpolarization in *Bacillus subtilis* leading to increased survival against spectinomycin, highlighting the involvement of membrane hyperpolarization in cell death [36]. It is possible that hyperpolarization of membrane might have led to enhanced killing. Aminoglycoside uptake is an energy-dependent process, requiring a high membrane potential [37]. ATP is produced by ATP synthase that is driven by a proton motive force (PMF) [38]. In the present study, increase in membrane potential by colistin might have led to increase in intracellular negative charge that might have increased the PMF and subsequently ATP production, which could have facilitated uptake of tobramycin.

The bactericidal effect of colistin has been shown to be mediated by ROS in *A. baumannii* [39]. Treatment of the late exponential phase *A. baumannii* cells with combination of tobramycin and colistin led to membrane hyperpolarization and production of ROS. This generation was attributed to colistin since treatment with tobramycin did not generate ROS. These results are in line with the study where high membrane potential has been shown to promote ROS production [40]. Membrane hyperpolarization has been associated with the

formation of superoxide radicals in *Escherichia coli* [41]. Tobramycin and colistin showed significant decrease in the expression of universal stress protein (*uspA*), which represents a family of conserved stress response proteins that help bacteria to adapt to oxidative stress, toxic metals, nutrient starvation, heat shock, DNA damage and antibiotic stress [42,43]. UspA has been shown to be important in combating oxidative stress in *A. baumannii* [44]. Hence reduced expression of *uspA* both by tobramycin and colistin might be responsible for enhanced lethality and eradication of persister cells.

Glyoxylate shunt, a bypass of the tricarboxylic acid (TCA) cycle that utilizes alternative carbon sources, is known to play an important role under several conditions including antibiotic defense and oxidative stress [45]. Inhibition of isocitrate lyase *icl*, a key enzyme of glyoxylate shunt, in *B. cenocepia* biofilms reduced survival of persisters against tobramycin [46]. Similarly, disruption of the *icl* gene decreased persistence and virulence of *Mycobacterium tuberculosis* in mice [47]. *icl* expression in *A. baumannii* was significantly down-regulated on exposure to tobramycin. However, treatment with colistin increased the expression of *icl*. Hence the role of *icl* in inhibition of persistence in *A. baumannii* against tobramycin-colistin combination was not clear.

Efflux pumps play an important role in tolerance to antibiotics. In *A. baumannii*, AdeABC efflux pump belonging to resistance nodulation-division (RND) and EmrAB efflux pump belonging to Major Facilitator Superfamily (MFS) have been shown to be responsible for the efflux of aminoglycosides and colistin, respectively [3,48]. A significant down-regulation of the efflux pump genes (*adeB*, *emrA* and *emrB*) upon exposure to tobramycin and colistin would also have contributed to the increased lethality and eradication of persister cells.

The present study shows that tobramycin in combination with colistin could eradicate persister cells, underlining the potential of this combination to effectively control *A. baumannii* infections. Since the combination eliminated persister cells completely, the underlying mechanisms including the production of ROS, enhanced membrane permeability and potential, and probably the sustained action of these antibiotics due to down-regulation of efflux pumps (Fig. 6), could only be studied with individual antibiotics.

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Figure legends

Fig. 1. Effect of antibiotic combinations on persister cells in late exponential phase of *A. baumannii* 17978 after treatment with (A) 40X tobramycin + 10X colistin, 40X tobramycin + 20X ciprofloxacin, and 40X tobramycin + 20X rifampicin, (B) 10X colistin + 40X rifampicin, 10X colistin + 20X ciprofloxacin, and 20X ciprofloxacin + 40X rifampicin, for different time intervals. Untreated cells (control) were taken as the 100% survival ($4.5 \pm 0.36 \times 10^9$ CFU/ml). The data is representative of three independent experiments. Bars represent the mean $\pm$ SD. **** $P \leq 0.0001$.

Fig. 2. Effect of 10X colistin, 40X tobramycin, and their combination on the outer membrane permeability of the late exponential phase cells of *A. baumannii* 17978. The data is representative of three independent experiments. Bars represent the mean $\pm$ SD. * $P \leq 0.01$; ** $P \leq 0.001$.

Fig. 3. Effect of 10X colistin, 40X tobramycin, and their combination on the membrane potential of the late exponential phase cells of *A. baumannii* 17978. The data is representative of three independent experiments. Bars represent the mean $\pm$ SD. * $P \leq 0.05$; ** $P \leq 0.001$; *** $P \leq 0.0001$

Fig. 4. ROS levels in late exponential phase cells of *A. baumannii* 17987 upon treatment with 10X colistin, 40X tobramycin, and their combination. The data is representative of three

independent experiments. Bars represent the mean $\pm$ SD. * $P \leq 0.05$, ** $P \leq 0.01$.

Fig. 5. Relative expression of universal stress protein *uspA*, isocitrate lyase *icl*, and efflux pump [*adeB*, *emrA*, *emrB*] genes, upon treatment with 40X tobramycin and 10X colistin in the late exponential phase cells of *A. baumannii* 17978. Untreated cells were taken as the control with basal level expression indicated as 1. The data is representative of three independent experiments. Bars represent the mean $\pm$ SD. * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$

Fig. 6. Proposed mechanism for the eradication of persister cells in *A. baumannii* by a combination of tobramycin and colistin. Colistin facilitated the entry of tobramycin into the cells and induced hyperpolarization resulting in increased ROS. Further, down-regulation of universal stress protein and efflux pumps by these antibiotics may be responsible for enhanced efficiency in combination (Created with BioRender.com).

Supplementary figure legend

Fig. S1. Persister cells in the late exponential phase of *A. baumannii* 17978 treated for 3 h with different concentrations of (A) tobramycin, (B) colistin (C) ciprofloxacin, and (D) rifampicin. Untreated cells (control) were taken as the 100% survival (4.4×10^9 , 3.9×10^9 , 3.3×10^9 and 3.8×10^9 CFU/ml, respectively). The data is representative of three independent experiments. Bars represent the mean $\pm$ SD.

Fig. S2. Persister cells in the late exponential phase of *A. nosocomialis* M2 treated with 40X tobramycin, 40X colistin, and in combination. Untreated cells (control) were taken as the 100% survival ($3.05 \pm 0.1 \times 10^9$ CFU/ml). The data is representative of three independent experiments. Bars represent the mean $\pm$ SD.

520

521 **Fig. S3.** Persister cells in the stationary exponential phase of *A. baumannii* 17978 treated for 3
522 h with different concentrations of (A) tobramycin, (B) colistin (C) ciprofloxacin, and (D)
523 rifampicin. Untreated cells (control) were taken as the 100% survival (5.6×10^9 , 4.9×10^9 , 4.2
524 $\times 10^9$ and 4.4×10^9 CFU/ml, respectively). The data is representative of three independent
525 experiments. Bars represent the mean $\pm$ SD.

526

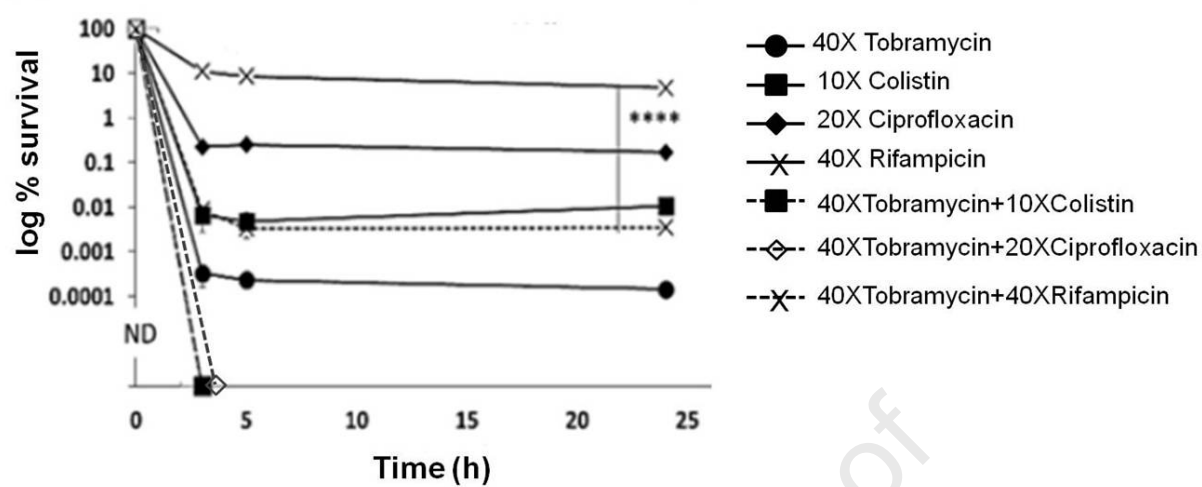
527 **Fig. S4.** Effect of antibiotic combinations on persister cells in stationary growth phase of
528 *A. baumannii* 17978 after treatment with 80X tobramycin in combination with 60X colistin,
529 60X ciprofloxacin, and 60X rifampicin, for different time intervals. Untreated cells (control)
530 were taken as the 100% survival ($5.0 \pm 0.28 \times 10^9$ CFU/ml). The data is representative of two
531 independent experiments. Bars represents the mean $\pm$ SD. $**P \leq 0.01$, $***P \leq 0.001$.

532

533 **Supplementary table caption**

534 **Table S1.** Primers used for gene expression analysis by qRT-PCR

(A)



(B)

